

PassportSahayak AI

Smart Passport & Consular Services Assistant

Software Requirements Specification (SRS)

Version 1.0 | Classification: Internal / Confidential
Powered by Spring AI | RAG-Based Knowledge Assistant

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional requirements for PassportSahayak AI — an intelligent, RAG-powered conversational assistant designed to handle all citizen passport-application queries and PSK/RPO official queries within the Passport Seva Programme.

PassportSahayak AI ingests structured passport policy documents, document-checklist annexures, police verification SOPs, fee schedules, and grievance/appeal procedures into a governed Knowledge Base (KB). All answers are retrieved from the KB, ensuring no hallucinated or policy-deviant responses reach applicants.

1.2 Scope

PassportSahayak AI serves two primary actors — Applicant and Official. The system scope for each is summarized below:

Applicant — General Passport Queries	Official — Passport Seva Assistance
- Passport eligibility validation - Fresh Passport applications - Renewal/Reissue applications - Tatkal guidance - Document checklist validation - Appointment slot search - Appointment booking - Appointment rescheduling - Appointment cancellation - Application status tracking - Grievance support	- Document verification support - Annexure verification support - Appointment capacity monitoring - Queue utilization monitoring - Police Verification workflow support - CCTNS process guidance - SLA monitoring - Fraud escalation guidance

OUT OF SCOPE: PassportSahayak AI does not grant or deny passports, override police verification findings, issue official passports, or make legally binding decisions on pending criminal/court cases. All disputed cases must be escalated to a PSK/RPO Official.

1.3 Document Conventions

Requirements use unique IDs: **FR** (Functional) and **NFR** (Non-Functional). Each requirement explicitly identifies its Actor as Applicant or Official. Priority levels: **Critical, High, Medium**.

2. System Overview

2.1 RAG Pipeline Flow

Applicant / Official Query → PII Redaction → Query Transformation → Vector Retrieval (MariaDB, threshold ≥ 0.5)-> Prompt Augmentation with Citations → LLM (Azure / gpt4+) → Grounded Response with Policy Citation

2.2 User Classes

Actor	Description	Primary Queries
Applicant	Citizen applying for a fresh, renewal, Tatkal, or reissue passport. Expects plain-language, policy-grounded answers.	Category eligibility, documents, PV status, appointment
Official	PSK/RPO staff supporting applicants and managing case load across categories and jurisdictions.	Annexure lookup, SOP reference, SLA tracking, escalation
Admin	KB administrators and operational administrators	-

2.3 Technology Stack

Component	Technology / Notes
Backend Framework	Spring Boot 3.x
AI Framework	Spring AI (latest stable) — RAG, embeddings, vector store
Chat LLM	Azure - gpt-4+
Embedding Model	text-embedding-3-small or ada
Vector Store	MariaDB(11.8+) with vector extension (cosine similarity search)
API Documentation	Springdoc OpenAPI (Swagger UI at /swagger-ui)
Language	Java 21+

3. Functional Requirements

The system exposes exactly 32 functional requirements, distributed across the supported actors. Each requirement is explicitly tagged with its Actor (Applicant, Official, or Admin) to enable role-based KB retrieval, access control, and audit logging.

3.1 Applicant Requirements (FR-001 – FR-015)

The following 15 requirements cover all core passport query domains an applicant may raise — from category selection and documents to police verification, appointments, and grievances.

Req ID	Actor	Requirement Description	Priority
FR-001	Applicant	The system shall determine eligibility for Fresh Passport applications using Knowledge Base rules.	Critical
FR-002	Applicant	The system shall determine eligibility for Renewal/Reissue Passport applications.	Critical
FR-003	Applicant	The system shall determine Tatkal eligibility and applicable service category.	Critical
FR-004	Applicant	The system shall provide passport validity, booklet type, and ECR/ECNR guidance.	High
FR-005	Applicant	The system shall provide document requirements for Minor Passport applications.	Critical
FR-006	Applicant	The system shall provide document requirements for Married Applicant categories.	High

Req ID	Actor	Requirement Description	Priority
FR-007	Applicant	The system shall provide Lost/Damaged Passport reissue requirements.	Critical
FR-008	Applicant	The system shall validate applicant document completeness before appointment booking.	High
FR-009	Applicant	The system shall retrieve available appointment slots based on PSK, POPSK, date, and application category.	Critical
FR-010	Applicant	The system shall allow booking of Fresh Passport appointments.	Critical
FR-011	Applicant	The system shall allow booking of Renewal/Reissue Passport appointments.	Critical
FR-012	Applicant	The system shall generate appointment acknowledgement and booking confirmation.	Critical
FR-013	Applicant	The system shall allow appointment rescheduling subject to slot availability.	High
FR-014	Applicant	The system shall allow appointment cancellation and display applicable refund policies.	High
FR-015	Applicant	The system shall provide application status tracking using ARN.	Critical

3.2 Official Requirements (FR-016 – FR-030)

The following requirements cover operational and administrative support for PSK front-desk staff, RPO back-office staff, Police Verification Cell users, and authorized Admin users responsible for governed KB ingestion.

Req ID	Actor	Requirement Description	Priority
FR-016	Official	The system shall provide end-to-end passport processing workflow guidance.	Critical
FR-017	Official	The system shall provide category-wise document and annexure verification support.	Critical
FR-018	Official	The system shall provide PV classification guidance (Pre-PV, Post-PV, PV-Exempt).	High
FR-019	Official	The system shall provide CCTNS dispatch and tracking workflow guidance.	High
FR-020	Official	The system shall provide PV turnaround and SLA monitoring support.	Medium
FR-021	Official	The system shall provide inter-state PV coordination guidance.	High
FR-022	Official	The system shall display appointment slot utilization and capacity by PSK/POPSK.	Medium
FR-023	Official	The system shall provide queue monitoring and appointment load management support.	Medium
FR-024	Official	The system shall provide cancellation, rescheduling, and no-show management support.	High
FR-025	Official	The system shall provide Citizens Charter SLA monitoring and breach identification.	Critical
FR-026	Official	The system shall provide grievance routing and escalation guidance.	Critical
FR-027	Official	The system shall provide appeal authority routing guidance.	Critical
FR-028	Official	The system shall provide fraud reporting procedures including CVC and ACB escalation guidance.	Critical
FR-016	Official	The system shall provide end-to-end passport processing workflow guidance.	Critical
FR-017	Official	The system shall provide category-wise document and annexure verification support.	Critical
FR-018	Official	The system shall provide PV classification guidance (Pre-PV, Post-PV, PV-Exempt).	High

Req ID	Actor	Requirement Description	Priority
FR-019	Official	The system shall provide CCTNS dispatch and tracking workflow guidance.	High
FR-029	Admin	The system shall support ingestion of passport circulars, annexures, SOPs, and operational documents into the Knowledge Base.	High
FR-030	Admin	The system shall maintain audit logs for KB ingestion, appointment transactions, and administrative activities.	High

4. System Constraints & Requirements

4.1 General System Constraints (Applicable to All Applications-Mandatory)

4.1.1 Knowledge Grounding

- All responses must be strictly grounded in Knowledge Base (KB) documents.
- No LLM-only or hallucinated responses are permitted.
- allowEmptyContext = false must be enforced globally — no response should be generated when KB context is unavailable.
- Retrieval similarity threshold must be ≥ 0.7 — low-confidence chunks must not be passed to the LLM.

4.1.2 Response Citation Requirement

- All responses must include citations referencing source Knowledge Base document and relevant section/page number.
- All responses must include the retrieved KB source details used to generate the answer.

4.1.3 Empty Context Handling

- If no relevant KB content is retrieved, the system must return the standard response:

```
{
  "answer": "No matching policy found. Please contact your nearest PSK/RPO.",
  "escalated": true
}
```

4.1.4 Session Traceability

- Every conversation session must have a unique UUID generated at initiation.
- Every retrieval request and generated response must be logged against that session UUID for end-to-end audit traceability.

4.1.5 KB Ingestion Response

- Upon successful document ingestion: HTTP Status 201 Created, with payload: documentId, chunks_created, ingestion_time_ms, metadata.

4.1.6 PII Detection & Redaction

- All Personally Identifiable Information (PII) must be detected and redacted before LLM processing.

PII Redaction Mapping

PII Type	Pattern Example	Redaction Token
Application Reference No. (ARN)	AA1234567890123	[ARN-REDACTED]
Passport Number	M1234567	[PASSPORT-NO-REDACTED]
Aadhaar Number	1234 5678 9012	[AADHAAR-REDACTED]
PAN Card Number	ABCDE1234F	[PAN-REDACTED]
Personal Phone Number	+91 98765 43210	[PHONE-REDACTED]
Personal Email Address	john.doe@gmail.com	[EMAIL-REDACTED]

PII Type	Pattern Example	Redaction Token
Home Address	123, MG Road, Bangalore 560001	[ADDRESS-REDACTED]
Date of Birth	15/08/1990 / Aug 15 1990	[DOB-REDACTED]

4.2 Passport Application-Specific Constraints

4.2.1 Compliance & Advisory Boundaries

- The system must not grant, deny, or guarantee a passport outcome, or override a police verification finding.
- All responses must be treated as informational and procedural guidance only.
- The system must not act as an authority for passport issuance decisions.

4.2.2 Escalation Requirement

- Any ambiguous, sensitive, dispute-related, or fraud-related query must be escalated to an Official using the standard escalation workflow.

4.2.3 Domain-Specific Safeguards

- The system must avoid interpreting policies beyond documented KB content.
- The system must avoid speculating on the outcome of a pending case or PV finding.
- The system must ensure neutral, compliant responses.
- The system must ensure strict adherence to Passport Seva Programme rules.

5. API Endpoints

Method	Endpoint	Description
POST	/ai/passport/chat/sync	Synchronous chat
POST	/ai/passport/chat/async	Streaming chat
GET	/appointments/slots	Slot search
POST	/appointments/fresh/book	Fresh appointment booking
POST	/appointments/renewal/book	Renewal/Reissue appointment booking
PUT	/appointments/{id}/reschedule	Reschedule appointment
DELETE	/appointments/{id}	Cancel appointment
GET	/appointments/{id}	Appointment details
GET	/applications/{arn}/status	Application tracking
POST	/kb/ingest	Knowledge ingestion

6. Glossary

Term	Full Form	Definition
RAG	Retrieval-Augmented Generation	AI technique combining vector retrieval with LLM generation for grounded, citation-backed responses.
KB	Knowledge Base	Repository of passport policy documents and SOPs used as the authoritative source for RAG retrieval.
LLM	Large Language Model	Azure / GPT-4+
PSK	Passport Seva Kendra	Front-end citizen service center for document verification, biometrics, and submission.

Term	Full Form	Definition
POPSK	Post Office Passport Seva Kendra	Passport service center operated through selected post offices to extend passport services.
RPO	Regional Passport Office	Back-end processing and issuing authority for passport applications.
PV	Police Verification	Verification of applicant identity, address, and record by the jurisdictional police station.
CCTNS	Crime & Criminal Tracking Network & Systems	National police network used to dispatch and track police verification requests.
ARN	Application Reference Number	Unique identifier generated for each passport application.
ECR/ECNR	Emigration Check Required / Emigration Check Not Required	Passport classification determining emigration clearance requirements before overseas travel.
Tatkal	—	Expedited passport scheme with reduced processing time and an additional fee.
PII	Personally Identifiable Information	Applicant data such as name, ARN, passport number, Aadhaar, PAN, phone number, email address, home address, and DOB.
CVC	Central Vigilance Commission	Apex vigilance and anti-corruption body of the Government of India that handles corruption and bribery-related complaints in central government organizations.
ACB	Anti-Corruption Bureau / Anti-Corruption Branch	Agency responsible for investigating corruption, bribery, and misconduct cases within the applicable jurisdiction.
CPV	Consular, Passport and Visa Division	A division under the Ministry of External Affairs responsible for passport, visa, and consular services; may serve as a higher appellate authority for passport-related appeals.
SOP	Standard Operating Procedure	Documented step-by-step instructions that define how a process should be carried out consistently.
NOC	No Objection Certificate	Official certificate issued by an authorized organization or employer indicating no objection to a passport-related request.
SLA	Service Level Agreement	Defined service timelines and performance commitments for passport processing and applicant services.
UUID	Universally Unique Identifier	Unique identifier used to track sessions, transactions, audit records, and system activities.